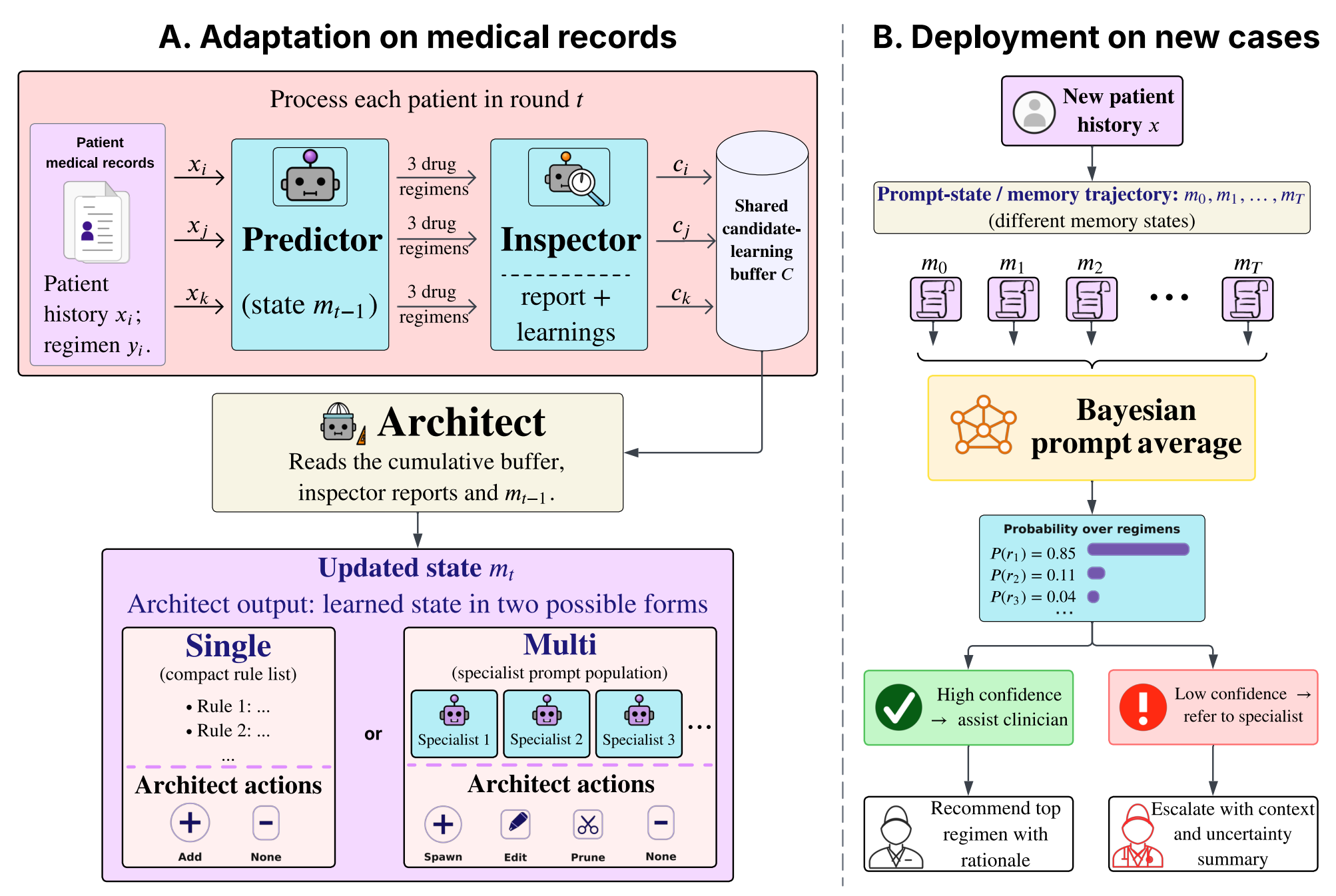

A. Adaptation on medical records
Process each patient in round t
Patient medical records
Patient history x_i; regimen y_i.
x_i
x_j
x_k
Predictor
(state m_{t-1})
3 drug regimens
3 drug regimens
3 drug regimens
Inspector
report + learnings
c_i
c_j
c_k
Shared candidate-learning buffer C
Architect
Reads the cumulative buffer, inspector reports and m_{t-1}.
Updated state m_t
Architect output: learned state in two possible forms
Single
(compact rule list)
• Rule 1: ...
• Rule 2: ...
...
Architect actions
Add
None
or
Multi
(specialist prompt population)
Specialist 1
Specialist 2
Specialist 3
Architect actions
Spawn
Edit
Prune
None
B. Deployment on new cases
New patient history x
Prompt-state / memory trajectory: $m_0, m_1, \ldots, m_T$
(different memory states)
m_0
m_1
m_2
m_T
Bayesian prompt average
Probability over regimens
$P(r_1) = 0.85$
$P(r_2) = 0.11$
$P(r_3) = 0.04$
High confidence → assist clinician
Low confidence → refer to specialist
Recommend top regimen with rationale
Escalate with context and uncertainty summary